

5.2 What materials and components should be selected when designing and manufacturing products and prototypes in Design Engineering?

- a. Understand that most products consist of multiple materials and that design engineers are required to discriminate between them appropriately for their use, including:
- i. ferrous, non-ferrous and alloy metals, such as:
 - mild steel, aluminium and brass.
 - ii. thermo softening and thermosetting polymers, such as:
 - HIPS, ABS and polyester resin, epoxy resin and polyimides.
 - iii. timbers and manufactured boards, such as:
 - oak, plywood and MDF.
 - iv. textiles used for reinforcement and coverings, such as:
 - geotextiles used in civil engineering and construction.
 - v. composite materials, such as:
 - fibre-reinforced plastics, glass-reinforced plastics (GRP) and carbon fibre (CFRP).
 - vi. smart materials, such as:
 - shape memory alloy, motion control gel, self-healing materials, thermochromic, photochromic and electrochromic materials.
 - vii. modern materials, such as:
 - sandwich panels, e-textiles, rare earth magnets, high performance alloys and super-alloys, graphene and carbon nanotubes.

